All-model F2/F3 trace, geometry and modular-order audit

Result. Kernel-check every general coefficient model over F2 and F3. For every nonsingular model, the actual source gcd(trace,q)≠1 predicate equals trivial geometric p-torsion. Actual Hasse inequalities and trace-square cases are proved here, with no Hasse/classification premise. Every actual point-count divisor r, including composite divisors, has gcd(r,q)=1 and modular order dividing4 over F2 or6 over F3. Generic actual point/equation-count semantics hold over every finite field. [pcc:unverified]

All F2 models	32 raw / 16 nonsingular / 8 source-SS models SS #E = 1, 3 or 5 → every count-divisor order | 4

All F3 models	243 raw / 162 nonsingular / 54 source-SS models SS #E = 1, 4 or 7 → every count-divisor order | 6

Actual enumeration and semantics. A coefficient-five-tuple equivalence covers every Weierstrass model. An actual nonsingular-point equivalence proves group cardinality equals the number of literal equation pairs plus the identity under nonzero discriminant; the actual integer trace equals the counted trace. Closed kernel tables prove all F2/F3 coefficient, point-count and Hasse cases. Counts include isomorphic coefficient duplicates. The earlier native 16+162 nonsingular-model baseline is credited; its execution is not promoted by these independent formal semantics. [pcc:unverified]

Source and geometric bridges. The source trace predicate is equivalent to a1=0 over F2 and b2=0 over F3. The preceding actual geometric j criterion identifies these with geometric killed-p subgroup cardinality1. Actual SS point-count cases derive the source trace-square alternatives. The older conditional modular-order arithmetic is credited; this packet supplies classification only over F2/F3. Root-of-unity powers give order-divisor bounds for every count divisor, without asserting every divisor occurs as a cyclic point order. General field-degree/full torsion-field bridges remain separate. [pcc:unverified]

Explicit positive controls. F2:$y^2+y=x^3+x$ has five actual rational points, P=(0,0) of exact order5, geometric two-torsion count1 and orderOf(2mod5)=4. F3:$y^2=x^3+2x+1$ has seven actual rational points, P=(0,1) of exact order7, geometric three-torsion count1 and orderOf(3mod7)=6. Both satisfy the actual source trace predicate and nonzero discriminant. Exact point orders use actual prime group cardinality/Lagrange. They lie outside the source p≥5 clause and do not refute it. [pcc:unverified]

Primary review. FST2010 pp230/237, Definition2.1/Remark2.2/Section3: exact trace predicate, embedding conditions, trace-square cases and p≥5 qualification retained. Milne, Elliptic Curves, IV9/Theorem9.4, printedp150: general Hasse proof uses degree/separability and is not assumed here. FST PDFpages7/14 and Milne PDF155 fully inspected; no full-paper/book read or corpus promotion. KB/Conductor unavailable, bounded fallback recorded. [pcc:unverified]

Validation and remaining scope. Whole gate44/0failures/0trusted native exceptions. Complete statement binding/recursive source-config pins. All83attempt-provenance files retained losslessly in32objects; five gate records, including failed default-limit attempts. Pinned Lean4.35.0/Mathlib56a00585…, only standard foundations. Exact local PCC replay differs from signed standing. General prime/extension fields, Hasse/endomorphism-rank/geometric identifications beyond this cohort, other BK primes/full all-k p<500 census and original fourteen-item gaps remain open. No novelty, qualified-source refutation, native experiment, paid compute, upload, holes/custom axioms/native evaluator/raised limits/new global flags/waivers. [pcc:unverified]